

ELEC 290: Final Report

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Statement of Originality

Our signatures below attest that this submission is our original work.

Following professional engineering practice, we bear the burden of proof for original work. We have read the Policy on Academic Integrity posted on the Faculty of Engineering and Applied Science website (<http://engineering.queensu.ca/policy/Honesty.html>) and confirm that this work is in accordance with the Policy.

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Executive Summary

The primary goal of this project is to design and engineer a problem proposed by the team that requires both software and hardware tools. The team ultimately chose to design and implement a Smart Bike Helmet (SBH). The SBH takes a traditional bike helmet and integrates a multitude of sensors and smart technology to enhance cyclist safety. As cycling accidents remain a significant safety concern, there was a need for a more comprehensive and proactive safety solution for cyclists.

The design incorporates various sensors, including accelerometers, force sensors, and light sensors, all coordinated using a software serial on an Arduino microcontroller. These components work together to detect crashes, measure distances, and improve visibility at night through integrated lights. The helmet also features a web server for data communication instead of a mobile app, ensuring accessible functionality while bypassing the challenges of app development.

The crash detection is a cornerstone of the design, utilizing accelerometers and force sensors to identify impacts and sudden movements. If a crash is detected, the system activates a response protocol. The helmet also includes distance detection to alert cyclists to nearby obstacles or vehicles, promoting situational awareness. Additionally, the light system enhances rider visibility during nighttime conditions, addressing a key safety concern for cyclists. With these key features, cyclists will be able to enjoy the modern functionalities without compromising the comfort or practicality of the helmet.

The development of the SBH was influenced by several key constraints. First, power efficiency was crucial, as the helmet needed to operate for extended periods on battery power without compromising functionality. Wireless connectivity was another major constraint, as the system required reliable communication between the helmet and external devices, such as a web server. Ensuring a stable connection over varying distances and under different conditions posed significant challenges.

If the project were to be redesigned, the team would explore using a printed circuit board (PCB) and a single microcontroller to streamline the design and improve efficiency. Weatherproofing the helmet for better durability in various conditions and developing a dedicated mobile app for seamless user interaction would also be priorities.

The Smart Bike Helmet aligns with growing demands for cyclist safety and convenience, offering practical and innovative solutions for modern riders. The project demonstrates the potential to significantly enhance cycling safety and accessibility.

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1. Introduction

While cycling offers a convenient, eco-friendly mode of transportation and exercise, it comes with its share of risks. As cities grow busier and more congested, cyclists face numerous hazards. From vehicle collisions to road obstacles, these can lead to serious accidents. Despite existing safety measures, such as helmets and reflective gear, there is a growing demand for more advanced and proactive safety solutions [1]. The purpose of the Smart Bike Helmet (SBH) is to address the core safety concerns of cyclists by integrating technology into a conventional bike helmet.

1.1 Background Research

From 1996 to 2016, the prevalence of cycling as a primary mode of transportation has doubled. This rise in cycling highlights the need for improved safety measures to ensure the well-being of cyclists on busy roads [2]. Statistics show that 85% Canadians believe that both motorists and cyclist must stay attentive and avoid distractions while navigating the roads [2]. When cycling, it is heavily advised to travel in the same directions as cars, if no bike lanes are available [3]. Therefore, it is important to be aware of one's surroundings, especially cars behind the cyclist. The Smart Bike Helmet uses distance detection to alert if a car is close to the cyclist.

When an emergency involving immediate danger or medical needs occurs, time is of the essence. The "golden hour", the first hour after an incident, rapid medical attention can be a matter of life or death [4]. For cyclists, who are particularly vulnerable to serious injuries during accidents, calling emergency services as quickly as possible can significantly increase the chances of survival. The SBH's crash detection system addresses this critical need.

1.2 Stakeholders

The development of the SBH involves several key stakeholders, each with specific needs and concerns. Cyclists, the primary users of the SBH, require a helmet that enhances their safety, provides convenience, and does not compromise comfort or usability. For them, the SBH must offer a seamless balance between functionality and ease of use.

Manufacturers and retailers also play a critical role as stakeholders. They are interested in a product that is cost-effective to produce, adheres to safety regulations, and has a strong market demand. Ensuring that the SBH is scalable for mass production is essential to gaining their support.

Lastly, emergency responders, although indirectly connected, benefit from the SBH's crash detection feature, which provides quicker access to critical information during accidents. This will allow them to improve their ability to offer timely assistance and can potentially save lives.

1.3 Problem statement

Cyclists face a significant safety risk due to various hazards on the road, including vehicle collisions, poor visibility, and accidents. While helmets and reflective gear offer some protection, they are largely reactive safety measures. The problem this project address is the lack of proactive, integrated safety technology that enhances cyclists' awareness of their surroundings and provides a comprehensive solution.

The SBH is guided by three main objectives: crash detection, nighttime visibility, and distance detection. The helmet's crash detection subsystem, powered by force sensors and an accelerometer, can detect when a user experiences a fall or impact, triggering an alert. This automatic response to potential injuries ensures faster assistance for cyclists who may be unconscious or unable to reach their phones. In addition to crash detection, the helmet provides real-time feedback about the cyclist's surrounding through the external readings' subsystem. With sensors to vehicle proximity, the SBH keeps the rider informed of environmental conditions that may affect their safety. The ultrasonic sensor is used to detect approaching vehicles from behind and the light sensor further assist in ensuring that cyclists are aware of potential hazards from all angles and all hours of the day.

The technical development of the SBH will involve a series of challenges, particularly in ensuring that all the sensors and components could be integrated into a helmet without compromising comfort or practicality. The project team conducted extensive research on sensor technologies, power management solutions, and communication protocols to ensure that the final product could meet the expectation of both safety and usability. The subsequent sections will outline the design concepts, development processes, and testing methods used in the project. Included figures will illustrate the helmet's design and the integration of electronic components, as well as a timeline of the project.

2. Design Process

Having thoroughly conducted background and stakeholder research into potential use cases and user needs, the team established a set of objectives, criteria, and specifications to guide the Smart Bike Helmet's (SBHs) development. This systematic approach allowed the team to create a framework to

inform the creative ideation process, ultimately enabling them to select the most promising design concept.

2.1 Objectives

There were three main objectives that the team thought was essential to the design of the SBH—starting with crash detection, where the helmet would have to be able to detect if the cyclist had been in an accident. This would be done with the primary goal of providing immediate emergency response to the user. To give this response, the SBH would send an alert to designated contact and emergency services, potentially saving lives. The second objective was to provide environmental monitoring to the user, letting the user know about their surroundings such as providing light when it got dark and detecting if a car was approaching from behind. Finally, global positioning system (GPS) notifications if the user arrived at their location was another objective to implement into the SBH system.

2.2 Design Criteria

The team developed five design criteria to ensure that the SBH's design was effective for the user: safety compliance, crash detection reliability, navigation system performance, ergonomic design, and battery performance. For safety compliance, the device would have to meet all existing safety certifications and be able to withstand significant impact forces, while providing enhanced emergency response. Then, for the reliability of the crash detection system, the algorithm developed would have to minimize false positive emergency alerts, include a user interaction timer to prevent unnecessary emergency calls, and be able to give accurate accident identification. Third, the GPS would have to provide real-time GPS updates with minimal latency and deliver clear alerts when the user arrives at their destination. Next was an ergonomic design, and for this, the helmet would have to maintain the comfort of a standard bicycle helmet, minimize the bulk and additional weight, and integrate the electronic components without compromising comfort. Finally, the battery performance of the device would have to support continuous operation for 8-12 hours and maintain continuous performance during extended cycling trips, while providing reliable power to all electronic subsystems.

2.3 Specifications

To achieve these objectives and criteria, the team would have to implement several sensors, communication modules, and a control system. In terms of sensors, force sensors, an accelerometer, and an ultrasonic sensor would be used for crash detection, while a light sensor would be used to detect

if it was dark outside. Two Arduino microcontrollers would power and control all components, while the GPS module would send location data to the Arduino and the wireless fidelity (WIFI) module would communicate information from the Arduino to the cloud. The Arduinos would be placed in a housing to protect them from damage and a breadboard would be used to route connections. Finally, an existing bike helmet would be the base for attachment for all these electronic components.

2.4 Creative Thinking

The Smart Bike Helmet (SBH) was an idea of a team member which was agreed on by the group. The decision of subsystems to be included in the design was a result of multiple mind maps, seen in Figure 1, using the SCAMPER¹ technique, and building off one another's ideas. The team made two lists of features for the helmet which were: must have features and bonus features. The next tool used was SCAMPER, to break down the project and features further. This tool helped the team to combine and eliminate some of the redundant features. Several points of SCAMPER seemed repetitive and did not fit the team's creative process; therefore, it was only used for the broader stage of the design process. The main ideas from the must have list were then mind mapped further to decide how to implement, and if implementation was possible. If implementation of these features seemed difficult, a creative solution was often required. If the creative solution was not feasible or the issue extended beyond the control of the team, the feature was disregarded. Features in the bonus list were mind mapped further and if deemed too complicated, or that they would take away from the main functionality of the helmet, they were set aside. If these features were simple and practical, they were broken down into steps and assigned to group members to accomplish.

¹ Substitute, Combine, Adapt, Modify (Also magnify and minify), Put to another use, Eliminate, and Reverse.

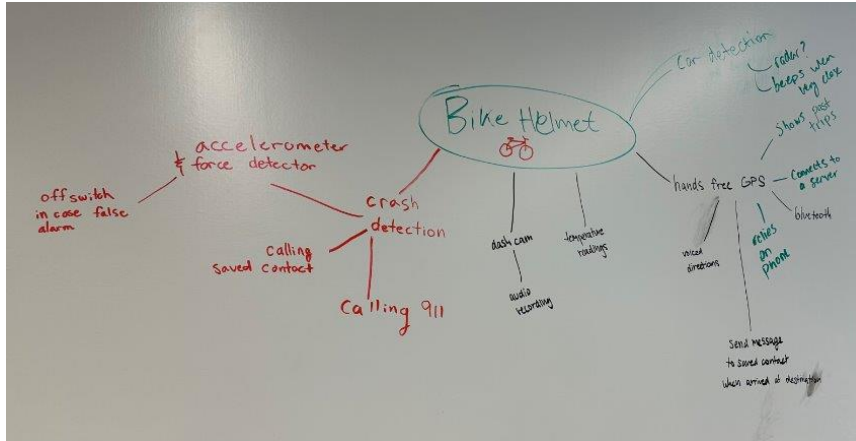


Figure 1: Mind map of SBH features.

3. Development of Solution

Faced with the problem of creating a bike helmet that can ensure the users safety and grant the user with a much more enjoyable experience, the team when through the full development process to assess the best solution that can provide maximum functionality to the user while adhering to the professional standards and requirements.

3.1 Concept Selection

Once the team had come up with desired subsystems and features, the team researched all features and made pros and cons lists. Research was done once most subsystems were determined, and this step helped refine the subsystems, and provide an understanding of what was possible in the project time period and with the resources provided. For example, after working on the crash detection system for a while, research proved that a call to a phone number (911 or other emergency service provider) would not be possible without a physical human interaction of pressing the call button. It was determined that code could be created to dial the phone number, but the physical call button had to be pressed by a human. This caused a change in the final design, to output a buzzer sound instead of a phone call. Pros and cons lists were heavily used in the design refinement process. Several lists were drafted for each subsystem, and this changed which subsystems were used and which were modified to create the Smart Bike Helmet.

3.2 Project Economics

The economic landscape of the Smart Bike Helmet (SBH) project was carefully navigated to ensure both technological innovation and financial feasibility. A comprehensive cost analysis revealed that the initial material expenses for the project were \$156, encompassing key components such as sensors (\$31), Arduino (\$45), communication hardware (\$40), and the base helmet (\$50), as seen in Table 1. When accounting for additional labour and 3D printing costs for the Arduino housing, the total project cost initially reached \$376.

However, strategic economic planning demonstrated significant potential for cost reduction. By leveraging bulk purchasing strategies, which typically offer 20-50% reductions from retail prices, and exploring labour outsourcing options that could lower expenses by 50-70%, the team projected the total project costs could be dramatically reduced to approximately \$130, as seen in Table 2. This approach exemplifies the team's commitment to economic efficiency without compromising technological quality.[1] [2]

Funding strategies were equally carefully considered, with multiple potential capital sources identified. University and federal grants targeting small businesses and sector-specific innovations represent important potential funding. Additionally, venture capital firms and angel investors could also be potential financial partners. Strategic partnerships with existing bicycle helmet manufacturers were explored as another way to lower base helmet costs.

Market analysis revealed that the LIVALL Smart Bike Helmet, retailing at \$203, would be the team's main competitor [7]. Through cost-reduction strategies, the team can potentially offer a competitive product with superior functionality at a substantially lower price point. The economic value proposition encompasses critical safety features such as accident detection, GPS tracking, and car detection. All of these are expected to provide tangible value to users through improved emergency response times and real-time safety updates.

Table 1: List of parts with associated costs.

Component	Description/Notes	Unit Cost (CAD)	Quantity
1. Bike Helmet	Pre-existing helmet for modification	\$50 - \$150	1
2. Arduino Uno Wi-Fi	Main microcontroller with Wi-Fi capability	\$45 - \$55	2
3. Ultrasonic Sensor (HC-SR04)	For car detection feature	\$5 - \$10	1
4. Accelerometer (GY521)	For crash detection and movement sensing	\$5 - \$10	1
6. Light Sensor (LDR)	Photo light-sensitive resistor for lighting detection	\$1 - \$5	1
7. Force Sensitive Resistor (FSR)	For detecting pressure or force on the helmet	\$10 - \$15	3
9. Buzzer (CMI-1295-0585T)	For audio alerts	\$5 - \$10	1
10. Breadboard	For prototyping and circuit assembly	\$5 - \$10	1
11. Wires & Connectors	Miscellaneous wiring for circuit connections	\$10 - \$20	1 set

Table 2: Total costs including parts and labour.

Category	Estimated Cost (CAD)
Parts and Components	\$156 - \$355
Manufacturing and Labour	\$220 - \$440
Estimated Total Cost	\$376 - \$795
Estimated Total Cost with Bulk Purchasing and Outsourcing	\$130 - \$350

3.3 Planning

The team met every Tuesday from 3:30-4:30 over the course of the semester to summarize upcoming tasks and ensure everyone was following the internal schedule set together at the start of the semester. During the meetings, work would be split up between group members and an internal due date would be agreed on to ensure everything was completed at a minimum one night before the due date. The plan and who each task was assigned to follow the GANTT chart as pictured in Figure 2, and the team was able to meet these deadlines effectively.

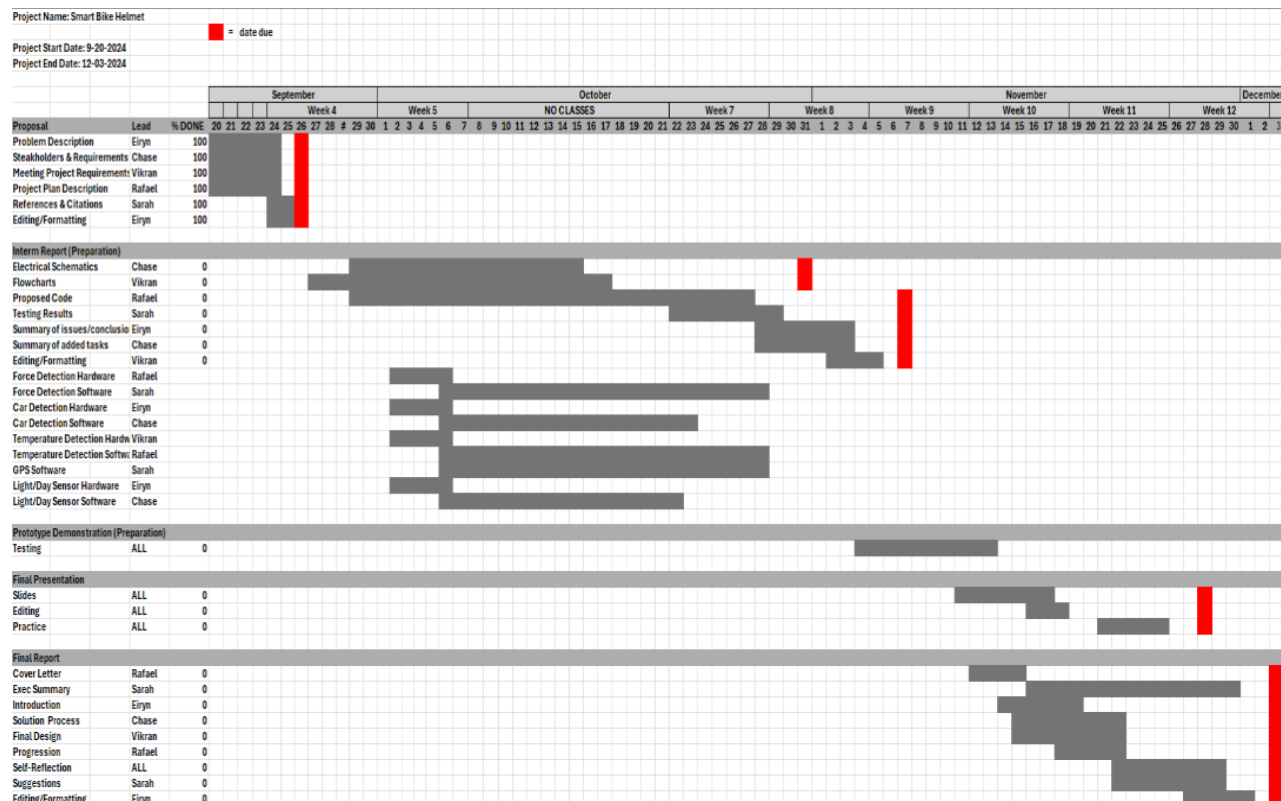


Figure 2: GANTT chart detailing order of task completion.

3.4 Environmental Considerations & Impact

Since traditional bicycle helmets are a one-time use product, the re-usability of the product was not an option from the start of the design process. The team decided to make the helmet out of three main modular parts, the helmet, the housing, and the wiring, so that in case of one part breaking or malfunctioning, only that one part would need to be replaced instead of the entire helmet. This is not only more financially friendly for users, but it also helps to reduce waste in terms of throwing out working components. Additional research into more sustainable and bio-degradable products which also

meet the safety standards for helmets could be done to ensure the smallest ecological footprint for the product. The use of bioplastics or plant-based materials would be ideal for helmet shells and padding as these materials biodegrade much faster, however lots of research and testing would need to be done with the safety standards to make sure the helmet maintains its functionality. In terms of the wiring components, many of these can be recycled and reused for new helmets, unless harmed in a crash. Even if components are harmed, they can most likely be recycled into new components which can then be used. One way to achieve this would be to establish a take-back program, where users would get a discount on a new helmet if they sent back their old helmet for recycling of parts. Other plastic components such as breadboard and wiring housing are most likely obsolete in a future design, reducing the need for them and therefore reducing the cost of purchasing them and the environmental effects which they cause.

3.5 Professionalism

When developing the SBH, many professional factors were taken into consideration including the safety of the helmet, the compliance of the design with current regulations, and the ethics of the design to the end user. The ethics of the design relied on three things: whether the user's privacy was infringed, the bias of artificial intelligence systems, and the decision harm and radicalization of the project. All of these were systematically integrated into the design process and there was a commitment from the team to professional engineering practices.

3.5.1 Safety

Being a safety device, the safety of the SBH was an integral part of the project as cycling helmets play a crucial role in protecting against head injuries, particularly traumatic brain injuries (TBIs) [8]. Making sure that the helmet still provided the biker protection during a bike ride was something the team wrestled with during the design process. Maintaining this safety was no easy feat as many components would have to be added to the helmet to introduce the SBH's features. Parts like the Arduino housings and the breadboard would introduce extra weight on the helmet and potentially reduce safety effectiveness. To accomplish this task, the team chose to use an existing bike helmet, instead of trying to build one from scratch, and would ensure that the baseline safety features of existing bike helmets would remain. From then, the team moved the Arduinos' housing to the back of the helmet, with the breadboard on top of the helmet to try and balance the helmet's weight.

3.5.2 Regulatory Compliance

Regulatory standards for helmets include many tests that the helmet must pass to be able to be sold to the public. For example, per the Consumer Product Safety Commission (CPSC) in the United States (which has standards used similarly in Canadian Law), the helmet is placed on an instrumented head form, turned upside down and dropped on an anvil from varying heights. The helmet then needs to register less than 300 g's^2 during the impact, and in some cases less than 250 or 200 g's [9]. The SBH would need to be put through these tests to become market viable, and to do so, the team designed the helmet to have all components securely strapped to the helmet using Velcro, hot glue, electrical tape, and having the Arduinos in a 3D printed compartment.

3.5.3 Ethics

Designing a device that all types of people will wear means considering a critical dimension of responsible engineering: ethics. This goes beyond just the technological specifications of the system, and how the project will impact the public socially.

3.5.3.1 Privacy

Privacy of the users' information is of utmost importance and was considered when developing the SBH. The Arduino does not send any sensitive information to the server when the device is running, and the only information shared is the rider's speed and the response of certain sensors needed for crash detection. Furthermore, the Arduino's server is locked under a password, which is only accessible to the team for test purposes.

3.5.3.2 AI (Artificial Intelligence) Bias

Although the SBH itself did not have artificial intelligence (AI) utilized in its components, AI was used to research and look at potential solutions for problems faced by the team. Therefore, any AI used in the development of the product was used with the awareness that the information is biased and was used with caution. Every result produced by an AI was reviewed by team members before being implemented into the design.

² $1 \text{ g} = 9.8 \frac{\text{m}}{\text{s}^2}$

3.5.3.3 *Decision Harm and Radicalization*

The team acknowledges that although a project might be made with the best of intentions in mind, there is always a possibility that there may be unintended uses and potential malicious uses of the technology. Potential avenues that the team predicts may be accessed by criminals are accessing the rider's GPS location. To mitigate this, the team decided to have the GPS module only alert the user when they had reached their location, instead of providing step-by-step directions that may be accessed, if the system were to be compromised.

3.6 EDII

Equity, Diversity, Inclusion, and Indigeneity (EDII) principles were fundamentally integrated into every stage of the design process, making sure people of all backgrounds were able to use and access the SBH. To start, as mentioned in Section 3.5.2, to meet regulatory compliance from the CPSC, tests must be run on the SBH, including using a head form. Utilizing both male, female, child, and adult head forms would allow the team to gain a more universal testing of the helmet and would help adapt the helmet to fit all age and gender groups. Also, since the team is using an already built bike helmet as a base for the other components, it is already proven that the helmet is utilized by all equity deserving groups. Finally, the pricing of the helmet, as mentioned in Section 3.2, is a big factor into whether customers will be able to access the product. To make it so it is more accessible to all groups, the team intends to use cost saving techniques to drive the market price of the helmet down.

4. Final Design

The finalised design of the Smart Bike Helmet encompassed the multiple subsystems that allowed maximum functionality, these subsystems were then compared to the project metrics to make sure the team met the requirements that were outlined. The final helmet design worn by a user can be seen in Figure 7.

4.1 Crash Detection System

The crash detection system has four main components: a GPS which can track the biker's location, an accelerometer which monitors acceleration and deceleration, force sensors which detect forces exerted on the helmet, and an ultrasonic sensor which can detect the distance between the helmet and an object behind it.

In real-life crashes, bikers are going to experience high forces on impact accompanied by large acceleration or deceleration. To prevent false positives, the crash detection system must distinguish between normal riding behavior and an actual crash.

For example, sudden deceleration alone could be a hard stop, and high forces alone could just mean something is leaning on a sensor, even if the values cross the threshold. Therefore, the logic in the code should detect crashes when multiple sensors provide a value over the threshold.

The Arduino will be updated constantly with values from the accelerometer, force sensors, GPS, and ultrasonic sensor. If the Arduino detects that acceleration and at least one force sensor crosses the threshold, it will output a loud sound through the buzzer. Only one force sensor is needed because a biker could only be hit on one side of the helmet, although the coding logic allows for multiple values to have a positive crash detection.

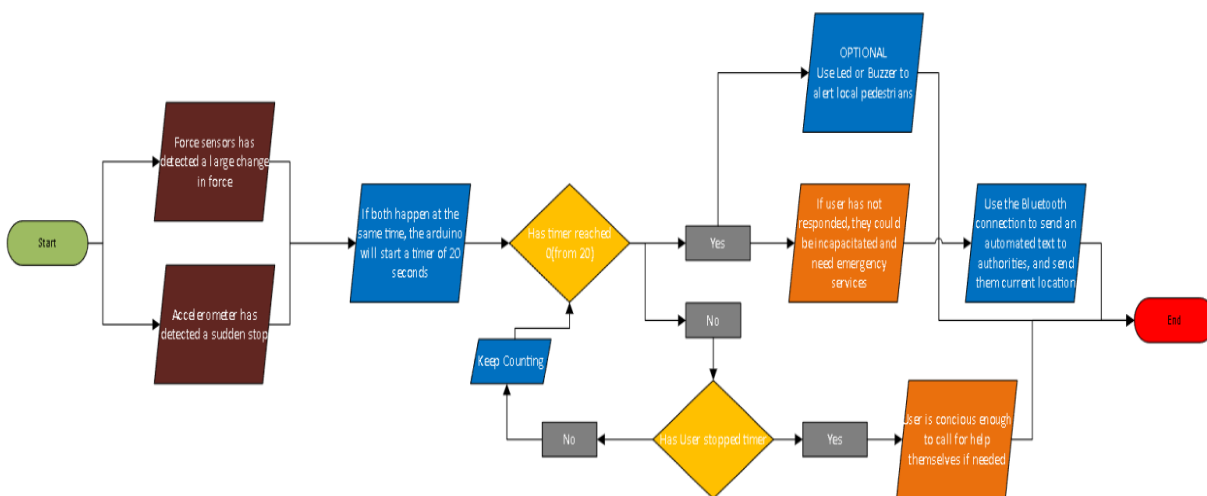


Figure 3: Logic of crash detection system.

4.2 Arduino Housing

The Arduino housing is a 3D printed structure which can hold two Arduino's, wiring, and electrical components in an organized and secure manner. Being a fully custom 3D printed structure, multiple features were designed to ensure the housing is stable, functional, and is easy to integrate into the prototype.

The bottom of an Arduino usually has metal protruding out of it that contributes to the overall functionality of the Arduino. This extra metal is non uniform, so if the Arduino was placed on a flat

surface, it would not be stable, and the fragile parts could snap off in a crash situation. Fortunately, the Arduino has holes in the board meant to fasten the Arduino to surfaces and this is what was used to make the socket pegging system. On the bottom of the housing, there are 4 half centimeters mounts that have holes extruded into them that line up with the holes on the Arduino. These mounts have more surface area than needed so when the board was being attached force wouldn't be concentrated on a small area and break it. Another large one was put near the middle of the board but had to be scrapped because the two boards were different, and it was incompatible.

Velcro Rails and the wedge were designed so the housing would have an easy way to be attached to the helmet. The rails are located on the bottom side of the housing and Velcro straps can be threaded through them. The Velcro was an easy solution to attach the hub to the helmet because it could be adjusted so the electrical systems could be modified but was still secure during crash testing. The wedge, when attached between the two Arduino compartments, made the bottom have an angle that would fit the top of a helmet better than a flat surface. All compact and functional helmets are usually round in design, so having a flat Arduino house wouldn't be practical in a final design.

A housing roof is a good solution to make sure hardware is protected against mild weather like rain or smaller debris like dust. The roof is friction fit and can slide into position securely without the need for additional attachments. The current design makes it easy to put on and take off whenever modifications in the interior are needed and makes it able to stay in place when using the helmet itself.

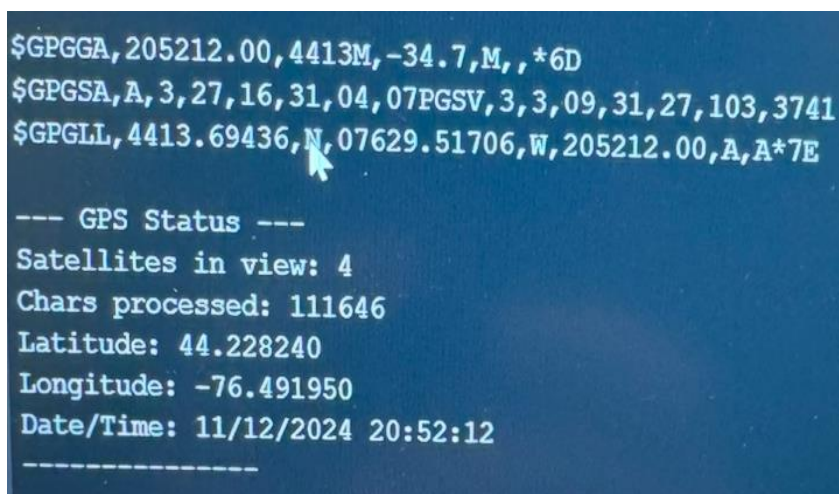
The front and the sides of the housing have slits which allow for wires to be routed through them. These openings are a good way for the Arduinos to communicate with each other without the wiring having to wrap around outside. It also acts as an access to the network of components stationed on the helmet. By routing the wires through the openings, good cable management can be achieved which can help with diagnosing issues with the helmet and overall understanding of the sub systems.

4.3 GPS System

The GPS subsystem was created using the GPS module that was connected to the Arduino, this module was able to communicate with satellites in space and gather readings for longitude and latitude of the user and output that data to the web server. The readings of the GPS module can be seen below in Figure 4. Some of the issues that came up with the current GPS module is that it had a long calibration time of approximately 20 minutes which made it difficult to use with the SBH, this is because for the GPS module to calibrate and provide accurate readings the module had to be stationary and not near any tall

structures so it could communicate with the satellites. This proves to be a problem because the GPS would be incompatible with users who might be riding bicycles in the city or if the user had trips less than 20 minutes, and in the end, it would be an inconvenience for users to not be able to use the GPS functionally of the SBH until it is done calibrating. A solution for this problem would be to upgrade the module, there are modules on the internet that have a quicker calibration time but do not sacrifice on size or power draw.

Future iterations of the GPS subsystem will see the implementation of a third-party mapping software which can be used to provide directions to the user and be used as a navigation tool.



```
$GPGGA,205212.00,4413M,-34.7,M,,*6D
$GPGSA,A,3,27,16,31,04,07PGSV,3,3,09,31,27,103,3741
$GPGLL,4413.69436,N,07629.51706,W,205212.00,A,A*7E

--- GPS Status ---
Satellites in view: 4
Chars processed: 111646
Latitude: 44.228240
Longitude: -76.491950
Date/Time: 11/12/2024 20:52:12
-----
```

Figure 4: GPS system response in Arduino console

4.4 Data Delivery

The next subsystem is data delivery this system focused on gathering the data that was collected from the sensors and outputting that data to the users.

4.4.1 Web Server

Part of the data delivery subsystem is the web server, which is used by the user to view the readings from the sensors on a webpage. This was accomplished by creating a web server using access point with the Wi-Fi Arduino. This allowed the user to create an access point to which they could connect with any device like a Wi-Fi connection, and when they visited the specific web address, they would be able to view the readings from the sensors as seen below in figure 5.

This method of connecting using the access point was chosen because of the connectivity benefits, with previous iterations the method of connection to the user was through a Wi-Fi connection, this came

with many disadvantages such as needing a Wi-Fi connection, and inconsistent connection. The chosen solution to this was to just use access point as it worked offline, providing all the same benefits as before with none of the Wi-Fi connection downsides.



Figure 5: Arduino to web server webpage screenshot.

4.4.2 Software Serial

The second part of the data delivery subsystem is software serial, this was used to communicate between the two Arduinos to send data from the slave Arduino to the master Arduino. This was accomplished by connecting both the Arduinos together as seen below in figure 6 and using software serial to communicate between both.

This method was chosen because of issues that came with different methods in previous iterations, one such issue is that the accelerometer and the GPS module was not detected when it was connected to the Wi-Fi Arduino, this meant that an additional Arduino was needed to run those two modules and the data from both would have to be sent back to the Wi-Fi Arduino. Another issue that arose with previous iterations is when the I2C method was used to try and communicate between both Arduinos, it became clear that the I2C chip in the Wi-Fi Arduino was not working as intended, this is possibly the reason why the accelerometer and GPS module did not function as intended as well. This is the reason

why the software serial method was the chosen form of communication between both the Arduinos as it did not rely on the I2C chip allowing it to transfer sensors data based of wired connection alone.

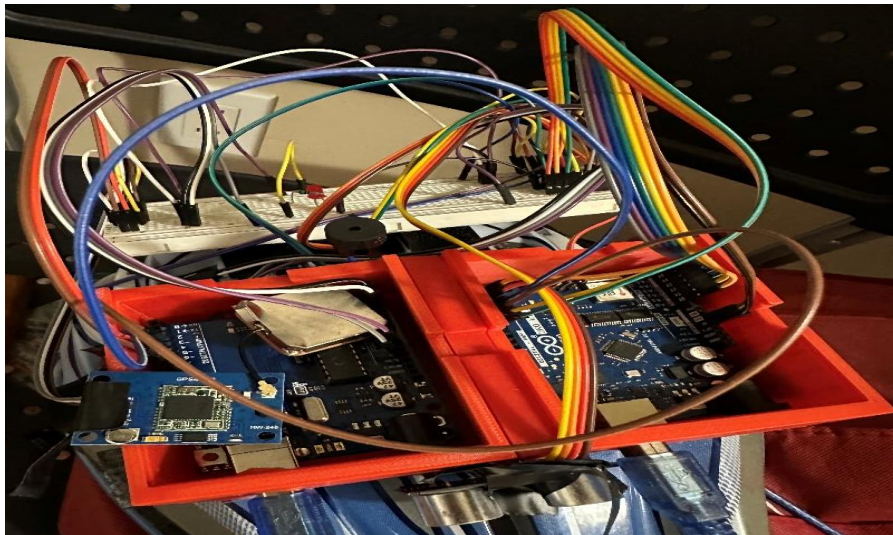


Figure 6: Two Arduino setup via Software Serial.

4.5 Light Sensor

Another functionality of the helmet is the light sensor, which works by detecting darkness, and turning a built-in light on. This feature was implemented by selecting a light threshold value, and when the sensor detected a value blow the selected value, it would turn on a LED on the helmet. Likewise, when the light value was detected to be higher than the selected value, it would turn off the light or keep it off if the light was not turned on yet. The use of this feature is when the user is biking at night, dusk, or dawn as it is important to be seen when it is darker out. This feature promotes safety for the user and provides a more enjoyable experience.

4.6 Distance Detection

The SBH's distance detection system represents a critical safety innovation, utilizing an ultrasonic sensor to provide real-time awareness of vehicles approaching from behind the cyclist. By emitting high-

frequency sound waves that bounce off nearby objects and measure return times, the sensor calculates the precise distance to approaching vehicles.

Specific distance thresholds were established to trigger different levels of alerts, ensuring that cyclists receive timely warnings when vehicles approach within a critically close range. Strategic sensor placement on the helmet was crucial to maximizing the detection range. The ultrasonic distance detection technology addresses a significant vulnerability for cyclists by extending awareness beyond natural visual capabilities. By creating an early warning system that can detect vehicles approaching from rear blind spots, the SBH offers a solution to one of the most critical safety challenges faced by cyclists on the road.

4.7 Assessment of Final Design to Project Metrics

The final design of the SBH demonstrated an achievement of the project's initial objectives across multiple domains: safety compliance, crash detection, navigation system, ergonomic design, and battery performance. Each subsystem underwent evaluation to ensure alignment with the team's established performance metrics and user requirements, as established in Section 2.2.

Safety compliance was prioritized through the selection of a standardized helmet that met existing protective equipment regulations. The crash detection system underwent extensive validation, with the team developing and implementing thresholds to accurately identify and respond to crash events. Upon detecting a crash, the system immediately activates a buzzer mechanism to signal the occurrence of a significant impact.

The navigation system presented the most substantial technical challenge, requiring significant scope refinement throughout the project. Initial GPS module limitations, including a long twenty-minute calibration period and unreliable third-party map integration, necessitated a strategic redesign. The team ultimately developed a modified approach that successfully displayed GPS coordinates on the server, representing a solution given the constraints of the available hardware.

Ergonomic considerations were carefully addressed through deliberate design choices focused on minimizing additional weight and maintaining user comfort. The team prioritized creating a helmet that not only provided technical functionality but also ensured a comfortable wearing experience. This approach recognized the critical importance of user experience in wearable technology design.

Battery performance was resolved through an innovative solution utilizing a portable power bank with appropriate adapters. This approach eliminated dependency on a fixed power source and enabled approximately eight hours of continuous helmet operation. The implementation demonstrated the team's ability to address power management challenges through creative engineering solutions.

While not every initial objective was fully realized in its original conception, the team successfully adapted and refined their approach to meet the project's core requirements. The SBH ultimately represented a balanced compromise between ambitious technical goals and practical implementation constraints, showcasing the team's engineering problem-solving capabilities.

5. Recommendations

Hardware optimization, enhanced software, and improved functionality are the recommended areas of focus for future iterations of the project. Focusing on these would ensure the Smart Bike Helmet's progress into a more effective solution by refining the reliability, usability, and performance of the bike helmet.

5.1 Hardware Optimization

In the current iteration, the solution is not very compact and has a clunky appearance. For all the implemented systems to work, two Arduinos were needed to spread the load from processing the different systems. An external breadboard was also needed to connect the different systems. In future iteration, using a more powerful microcontroller would mean only one would have to be used, which would cut the Arduino house in half. Alternatively, a smaller microcontroller would be a decent substitute if two microcontrollers were still needed. To connect the systems, a custom PCB could be designed with space optimization in mind. The breadboard limited how optimized the space could be, as only half the board was ever used. If the breadboard could have been cut in half, the functionality would have stayed the same, but the overall appearance would have improved. Another way to decrease the size is to integrate a battery with a slim profile onto the helmet. In the current iteration, a large portable battery was needed to power the helmet, but a slim battery would reduce the size and weight by considerable margin.

Lastly, in any future iterations durability and weather resistance should be accounted for when designing the hub and choosing hardware. Part of the safety considerations when making the current iteration was making a hub that could reliably hold the Arduinos and protect them during testing, but in

future iterations where the helmet will be outside protecting the electrical components from weathering is crucial because that poses another safety concern.

5.2 Software Development

In future iterations the team would focus on mobile app development which could connect to the Smart Bike Helmet would allow for better data transferring and allow the helmet to use that data more effectively. The app would be a user-friendly way for the user to customize their own experience and would allow for many sub-systems to be implemented.

One of the main features of the app would be transferring data for automatic alerts when crashes are detected. Currently, the location can be stored on the Arduino, but it is not able to be used. If an app was developed, all the data could be stored in a mobile phone and be user-friendly to access. The app could then be used to transfer data like location, emergency contact information, or other relevant information of the user when implementing things like the automatic messaging system.

Another main feature would be customization, where users could customize their thresholds. There are many different types of biking, like mountain biking, racing, or regular biking to work. There is also the chance that the Smart Bike Helmet can become a universally smart helmet and be implemented for motorcycles. All these different types of riding will have different crash scenarios, so the design needs to be able to be customizable so users can change it to their own needs.

Lastly, hands-free GPS would allow users to navigate to their destination without being unsafe while looking at their phone. In the current iteration, the Arduino gets data about the current location in the form of longitudinal and latitudinal coordinates but cannot use them to give directions. If an app was developed those coordinates could be passed through another map app and then directions could be played through a speaker. This would increase the amount of time a biker's eyes are on the road, which would allow them a better understanding of any dangers on the road. The next iteration would aim to reflect a real-world solution, where the design offers a reliable and user-friendly system which is capable of immediate support in a worst-case scenario.

5.3 Crash Detection Refinement

Another focus for future iterations of the project is an overall refinement of the crash detection system. Since crash detection is the focus of the project and the main concern of the stakeholders improving it is a top priority. There are many ways to do this, although some recommended secondary focuses should

be on finishing the automatic alert feature, extensive testing and adjustments to coding logic, and continuing to develop pre-crash alert systems.

In the current iteration, when a crash occurs, the Arduino will output to a buzzer located on the top of the helmet and although that might be good for getting the attention of people in the vicinity a more realistic solution is needed. By using an app, the Arduino could connect and utilize the user's mobile phone via Bluetooth. If a crash was detected, the system could automatically send a text message to authorities and emergency contacts. These messages would have information like the location and severity of the detected crash. The app could also implement a timer for a false alarm feature which gives the user time to cancel the automatic message if it was a false alarm. For future iterations, more broad testing should be done to test the coding logic with any edge cases when it comes to crashes. The coding logic right now is very effective for a normal crash situation but may not be effective under certain circumstances. For example, if a biker is hit very hard on the head but they do not slow down enough to cross the threshold in the coding logic then the system might not automatically message even if the biker is hurt. A possible solution could be that the Arduino stores values for a timeframe and if the thresholds are crossed within that time frame, it could output a message. Lastly, the team implemented an ultra-sonic sensor on the back of the helmet to warn if any objects came near the user. If more sensors were used in the blind spots and a suitable distance was found through testing, the system could act as early warning and crash prevention. This could alert the rider to possible dangers behind them and allow them time to take evasive action, potentially saving them from a crash.

5.4 Modular Realisation

The last focus is something that addresses the main problem with bike helmets, they are ultimately single-use products once they have been in a crash scenario. In the current iteration of the design, the hardware components are integrated into the helmet and are essentially part of the helmet. This means that if a crash happens the hardware could be damaged beyond repair along with the helmet, which means the hardware would also essentially be single use. This means that this product might seem less financially feasible in the long run, and recyclable materials would have to be prioritized because there would be a lot of waste after the helmet is damaged. Therefore, in future iterations, the design implementation would be shifted from a fully integrated helmet to a universal helmet attachment which has the same capabilities. This attachment could be made using microcontrollers and custom PCBs. The attachment would be placed on a helmet where it has the least risk of impact and would allow the user to transfer it to a new helmet if a crash happens.

6. Conclusions

In conclusion, through idea creation techniques like SCAMPER, mind mapping, and building off other ideas the group was able to come up with many concept solutions associated with the Smart Bike Helmet. After doing extensive research and comparing all possible solutions with the project scope, multiple subsystems like crash detection, distance detection, GPS, and data delivery through a web server were chosen to be implemented into the final design. The final design implemented subsystems within functional requirements, but the physical design was not as compact or efficient as what the team would have liked. The team has noted multiple focusses that the project should have if it is continued. Implementation of smaller, more powerful, or customized components will allow for a more compact and streamlined design. The development of an app could make subsystems that were previously excluded more feasible. Lastly, further development of the crash detection system and addition of the automatic messaging system would reflect real life crash scenarios more. Overall, the project was deemed a success with room for improvement in future iterations.

7. Appendix

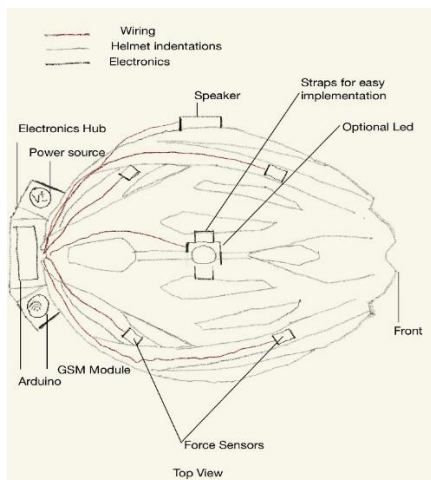


Figure 7: Detailed Sketch of Initial Helmet Design.

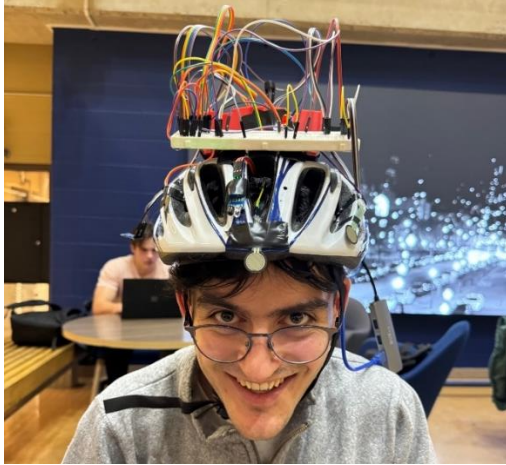


Figure 8: User Wearing Final Design of Helmet.

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